

Lesson 06: Solving Basic Differential Equations with Integration

Professor Leonard Differential Equations Master Series · Lesson 6

- **Direct Video Link:** [Lesson 06: Solving Basic Differential Equations with Integration](#) · Duration: 39:21
- **Target Exam Scope:** 🎯 MIDTERM EXAM (Ch 1.2 — Direct Quadrature & Constant Management)
- **Course Context:** Applied Ordinary Differential Equations (ENGR 213)

1. Whiteboard Lecture Intuition & Philosophical Framing

In this 39-minute lecture, Professor Leonard dives into the simplest class of differential equations: direct quadrature equations of the form $dy/dx = f(x)$. While conceptually straightforward, this lecture highlights the absolute necessity of rigorous integration constant management, tracking signs, and handling transcendental and rational antiderivatives.

2. Core Theoretical Concepts & Mathematical Formulations

Direct Quadrature Form: $dy/dx = f(x)$

When the derivative depends strictly on the independent variable x and contains no dependent variable y :

$$\frac{dy}{dx} = f(x) \implies dy = f(x)dx \implies y(x) = \int f(x)dx + C$$

The 3 Crucial Rules of Direct Quadrature:

1. **Never Drop $+C$:** The integration constant is part of the antiderivative, not an add-on.
 2. **Review Fundamental Integration Techniques:**
 - u -substitution: $\int g(u)u'(x)dx$.
 - Integration by parts: $\int u dv = uv - \int v du$.
 - Partial fractions: Decomposing rational functions $P(x)/Q(x)$.
 - Trigonometric substitutions and trigonometric identities.
 3. **Repeated Quadrature for Higher-Order Equations:**
 - $y'' = f(x) \implies y'(x) = \int f(x)dx + C_1 \implies y(x) = \int y'(x)dx + C_2$.
 - Each integration produces an additional independent constant.
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3. Classroom Chalkboard Examples (Solved in Complete Baby Steps)

Problem 06.1: Chalkboard Problem 6.1: Direct Quadrature with Integration by Parts

Problem Statement:

Solve the initial value problem: $\frac{dy}{dx} = xe^{-2x}$, subject to $y(0) = 1$.

Step-by-Step Whiteboard Solution:

- **Step 1: Separate Differentials and Set Up the Indefinite Integral:**

$$dy = xe^{-2x} dx \implies y(x) = \int xe^{-2x} dx$$

- **Step 2: Apply Integration by Parts (IBP):** Let $u = x \implies du = dx$ Let $dv = e^{-2x} dx \implies v = -\frac{1}{2}e^{-2x}$

$$\int u dv = uv - \int v du$$

$$y(x) = x \left(-\frac{1}{2}e^{-2x} \right) - \int \left(-\frac{1}{2}e^{-2x} \right) dx$$

$$y(x) = -\frac{1}{2}xe^{-2x} + \frac{1}{2} \int e^{-2x} dx$$

$$y(x) = -\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x} + C$$

- **Step 3: Apply the Initial Condition $y(0) = 1$ to Determine C :**

$$1 = -\frac{1}{2}(0)e^0 - \frac{1}{4}e^0 + C$$

$$1 = 0 - \frac{1}{4}(1) + C \implies 1 = -\frac{1}{4} + C \implies C = \frac{5}{4}$$

- **Step 4: State the Particular Solution:**

$$y(x) = -\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x} + \frac{5}{4} = -\frac{1}{4}e^{-2x}(2x + 1) + \frac{5}{4}$$

- **Step 5: Differentiate to Verify:**

$$y'(x) = -\frac{1}{4} [-2e^{-2x}(2x + 1) + e^{-2x}(2)] = -\frac{1}{4}e^{-2x} [-4x - 2 + 2] = -\frac{1}{4}e^{-2x}(-4x) = xe^{-2x} \quad \checkmark$$

$$y(0) = -\frac{1}{4}(1)(1) + \frac{5}{4} = \frac{4}{4} = 1 \quad \checkmark$$

[!WARNING] **Common Exam Pitfall:** Sign errors during integration by parts. Tracking the double negative $-\int v du = -\int (-1/2e^{-2x})dx = +1/2 \int e^{-2x} dx$ is the most frequent student stumbling block.

4. Key Takeaways & Exam Strategy Formula Card

- Watch the full whiteboard demonstration on YouTube: [Lesson 06: Solving Basic Differential Equations with Integration](#)
- Master the classification and verification mechanics before attempting complex integrations.
- Always check domain restrictions and verify initial values back into the computed trajectory.